



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

AI for IT Security Professionals

TGS Ref No: TGS-2023039344

Conducted by

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Version v6.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v5	Legacy reference	Previous Azure course deck and supporting material	Tertiary Infotech
v6.0	6 September 2026	Full AI security rebuild; 16 self-contained labs; preserved live assessment structure	Tertiary Infotech

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Course delivery and alignment

AI for IT Security Professionals | TGS-2023039344 | v6.0. 32 contact hours: 15 facilitated, 15 practical, 2 assessment. Breaks excluded. The slide map below is generated from the current deck.

LO1: Administer security programmes and analyze the impact of system updates.

LO2: Perform system administration and configure network device security features.

LO3: Perform troubleshooting of security software and assist users in defining access rights.

LO4: Coordinate access control rights and investigate unauthorized access incidents.

Day 1

Contact time 480 minutes. Training 480 minutes.

Lab	Topic and LO	Slides	Facilitated	Practical
1	Risk register and patch rollout LO1	15–38	60 min	60 min
2	Redaction and AI evidence validation LO1	39–62	60 min	60 min
3	RBAC and privileged access LO3	64–87	60 min	60 min
4	Conditional Access and workload identity LO3	88–111	60 min	60 min

Facilitation method: inspect the mechanism and field contract, challenge the counterexample, and explain the decision logic. Practical method: run the baseline, change one fixture, validate an AI claim and capture operational evidence or a precise limitation.

Resources: current visual deck, Learner Guide, the four named lab folders, Python 3.10+, and the assigned Azure sandbox. Use the lab acceptance checks for formative feedback. Do not treat an offline pass as a deployed control.

Suggested break placement: 15 minutes after the first block, 60 minutes lunch after the second, 15 minutes after the third. These 90 minutes are outside the 480 contact minutes; adjust clock times to the confirmed class timetable.

Day 2

Contact time 480 minutes. Training 480 minutes.

Lab	Topic and LO	Slides	Facilitated	Practical
5	NSG segmentation and flow diagnosis LO2	114–137	60 min	60 min
6	Web edge and private access LO2	138–161	60 min	60 min
7	VM administration and encryption LO2	163–186	60 min	60 min
8	Storage SQL and secret protection LO2	187–210	60 min	60 min

Facilitation method: inspect the mechanism and field contract, challenge the counterexample, and explain the decision logic. Practical method: run the baseline, change one fixture, validate an AI claim and capture operational evidence or a precise limitation.

Resources: current visual deck, Learner Guide, the four named lab folders, Python 3.10+, and the assigned Azure sandbox. Use the lab acceptance checks for formative feedback. Do not treat an offline pass as a deployed control.

Suggested break placement: 15 minutes after the first block, 60 minutes lunch after the second, 15 minutes after the third. These 90 minutes are outside the 480 contact minutes; adjust clock times to the confirmed class timetable.

Day 3

Contact time 480 minutes. Training 480 minutes.

Lab	Topic and LO	Slides	Facilitated	Practical
9	Policy and security posture LO1	213–236	60 min	60 min
10	Security logs and detection metrics LO4	237–260	60 min	60 min
11	Access rights troubleshooting LO3	262–285	60 min	60 min
12	Unauthorized access investigation LO4	286–309	60 min	60 min

Facilitation method: inspect the mechanism and field contract, challenge the counterexample, and explain the decision logic. Practical method: run the baseline, change one fixture, validate an AI claim and capture operational evidence or a precise limitation.

Resources: current visual deck, Learner Guide, the four named lab folders, Python 3.10+, and the assigned Azure sandbox. Use the lab acceptance checks for formative feedback. Do not treat an offline pass as a deployed control.

Suggested break placement: 15 minutes after the first block, 60 minutes lunch after the second, 15 minutes after the third. These 90 minutes are outside the 480 contact minutes; adjust clock times to the confirmed class timetable.

Day 4

Contact time 480 minutes. Training 360 minutes plus 120 minutes assessment.

Lab	Topic and LO	Slides	Facilitated	Practical
13	Prompt injection and retrieval boundaries LO4	312–335	45 min	45 min
14	Tool gateway and response approvals LO4	336–359	45 min	45 min
15	AI security evaluation LO1	361–384	45 min	45 min
16	Incident response and service recovery LO4	385–408	45 min	45 min

Facilitation method: inspect the mechanism and field contract, challenge the counterexample, and explain the decision logic. Practical method: run the baseline, change one fixture, validate an AI claim and capture operational evidence or a precise limitation.

Resources: current visual deck, Learner Guide, the four named lab folders, Python 3.10+, and the assigned Azure sandbox. Use the lab acceptance checks for formative feedback. Do not treat an offline pass as a deployed control.

Assessment: Written Assessment 60 minutes (9 questions, K1–K9); Practical Performance 60 minutes (3 tasks, A1–A6). Individual, open book. Submit to LMS; assessor records C or NYC and handles clarification without coaching.

Suggested break placement: 15 minutes after the first block, 60 minutes lunch after the second, 15 minutes after the third. These 90 minutes are outside the 480 contact minutes; adjust clock times to the confirmed class timetable.

Assessment mapping

K1 → WA question 1 → RBAC and PIM → Labs 3, 11 → Slides 64–87, 262–285

K2 → WA question 2 → Conditional Access and workload identity → Labs 4 → Slides 88–111

K3 → WA question 3 → NSG ASG and flow diagnosis → Labs 5 → Slides 114–137

K4 → WA question 4 → Firewall WAF and Private Endpoints → Labs 6 → Slides 138–161

K5 → WA question 5 → Bastion JIT and encryption → Labs 7 → Slides 163–186

K6 → WA question 6 → Storage SQL and Key Vault → Labs 8 → Slides 187–210

K7 → WA question 7 → Policy and Defender → Labs 9 → Slides 213–236

K8 → WA question 8 → Sentinel investigation and response → Labs 10, 12, 16 → Slides 237–260, 286–309, 385–408

K9 → WA question 9 → Evidence limitations and validation → Labs 1, 2, 15 → Slides 15–38, 39–62, 361–384

A1,A2 → PP Task 1 → Labs 3,4,11. A3,A4 → PP Task 2 → Labs 5,6,7,8. A5,A6 → PP Task 3 → Labs 9,10,12,16. This preserves the live legacy paper's item tags; LO labels are supplemental teaching alignment.